

Introduction and Transmission Guide

DBA-MA Oil and Natural Gas Series

Series Overview Document

Design Basis Accident — Military Action Framework

Version 1.0

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Preface

In February 2022, I wrote a letter warning that Russian military action against Ukrainian energy sector facilities was imminent. The invasion came days later. Zaporizhzhia — Europe’s largest nuclear plant — was captured at gunpoint, and the world discovered that I was right about the gap. Upon reflection over the events just after the attack, I came to realize that the international nuclear safety framework had no methodology for assessing facility safety under an active military threat.

That letter became the origin of the Design Basis Accident — Military Action framework. The DBA-MA work began in the nuclear sector because that was where the gap was most visible and the consequences most severe. But the logic of the framework — defining the bounding military-action threat that critical infrastructure must be hardened against — applies across every sector where a deliberate military attack is a credible threat.

The Oil and Natural Gas sector is such a sector. Nord Stream proved it. The Russian campaign against Ukrainian energy infrastructure proved it again. The Colonial Pipeline incident proved that even a criminal ransomware attack against a voluntary-compliance sector produces consequences that a nation-state adversary would be delighted to achieve deliberately. And the formal CIKR designation of the sector — the government’s own statement that its disruption would have national security consequences — proved that everyone already knew this and had chosen not to act on that knowledge.

This document introduces the three-document Oil and Natural Gas series of the DBA-MA framework. It is written primarily for Kimberly Denbow and for the peers across the sector’s trade associations who will receive this material. But it is also written as a guide for the conversations that need to follow — with member companies, with multinational leadership, with congressional staff, and with the regulators who have the authority to close the gap that the voluntary framework has left open for too long.

Seven visits to Ukraine informed the parent framework. Thirty years in critical infrastructure security — including the development of the mandatory framework that protects the electricity sector — inform this series. The gap is real. The threat is demonstrated. The pathway to closing it is available. This document lays out who needs to walk it and how.

1. The Document Set

The DBA-MA Oil and Natural Gas series consists of three documents, each serving a distinct analytical purpose and a distinct audience. They are designed to be read as a set, but each stands on its own. The following characterizations tell you what each document does and who benefits most from reading it.

1.1 The Child Document: DBA-MA Oil and Natural Gas Sector (V1.1)

The Child Document is the technical foundation of the series. It applies the DBA-MA methodology — the nuclear safety concept of a Design Basis Accident, extended to the deliberate military-action threat tier — to the Oil and Natural Gas sector’s specific asset profile, operational characteristics, and regulatory environment.

The document defines three bounding scenarios: a coordinated attack on interstate natural gas compressor stations (Scenario A), a refinery cyber-kinetic sequence (Scenario B), and combined pipeline rupture and ignition (Scenario C). For each scenario, it traces the initiating event, the progression sequence, the consequence envelope, the protective action window, and the recovery timeline. It establishes the simultaneity trap — why scenarios that are individually survivable become jointly unrecoverable when executed in coordination — and identifies the regulatory, hardening, information-sharing, and R&D gaps that the current framework leaves unaddressed.

The Child Document also grounds the threat in the evidentiary record. Nord Stream I and II — as proof of concept for coordinated pipeline rupture as military action. The Russian campaign against Ukrainian energy infrastructure is the most extensively documented operational execution of a sustained attack on natural gas compression, storage, and generation at a national scale.

Read this document first if you are: a facility security officer or operations professional responsible for pipeline, refinery, or compressor station security planning; an exercise designer developing sector-wide scenario-based preparedness events; or a government official responsible for sector risk assessment or emergency response planning.

1.2 The Use Case: LNG Terminal Attack (V1.0)

The LNG Use Case is the strategic supply chain document of the series. It applies the DBA-MA methodology to a specific and consequential attack vector: deliberate military action against liquefied natural gas terminal infrastructure. Written for a dual U.S./NATO audience, it addresses both domestic terminal security and the alliance-level energy security implications of terminal disruption.

The Use Case leads with strategic consequence rather than physical consequence. The BLEVE and vapor cloud explosion envelope — the immediate physical safety consequence of an LNG release — is held for a future companion document. This document focuses on what matters most to the NATO audience: what happens to allied energy security, European winter heating supply, and military logistics fuel chains when U.S. export terminal capacity is degraded or destroyed, and what that means for alliance cohesion under the adversary’s strategic design.

The document addresses all three UAS modalities — aerial, maritime surface, and subsea — as named attack vectors, using the Houthi maritime campaign and the Nord Stream incident as evidentiary anchors. It profiles the strategic terminal inventory from Sabine Pass to Świnoujście. It identifies the USCG/FAA counter-UAS authority gap and the FSRU security gap created by the rapid post-2022 European deployment.

Read this document first if you are: a NATO energy security planner, a European government official responsible for LNG import security, a multinational energy company executive with U.S. export and European import assets, or a maritime security practitioner.

1.3 The Policy Companion: Deterrence Without Standards (V1.1)

The Policy Companion is the legislative instrument of the series. It makes the case for mandatory physical security standards for the Oil and Natural Gas sector — the CIP-014 equivalent that the electricity subsector has, but that the Oil and Natural Gas sector lacks — using deterrence theory as the governing argument rather than a regulatory comparison.

The document opens with the CIKR logical predicate: the federal government has formally designated the sector as Critical Infrastructure and Key Resources, which means its disruption has national security consequences, which means attacks against it are national security threats, which means managing those threats with voluntary guidelines is internally inconsistent with the government's own position. That argument is the thesis. Everything that follows — the Colonial Pipeline case study, the CIP-014 comparison, the legislative vehicles, the trade association business case — develops and supports it.

The document makes a direct and specific ask to the sector's trade associations, organized around a three-part offer: liability protection for compliant operators, cost recovery through rate cases for regulated entities, and collaborative standard development modeled on the NERC/FERC process. It addresses the full trade association landscape — AGA, API, INGAA, AXPC, IPAA, and NGSA — and makes the case for multinational company leadership as an independent pathway that complements rather than replaces the legislative track.

Read this document first if you are: a trade association executive, congressional staff member, TSA or FERC official, multinational company C-suite or government affairs officer, or any reader whose primary interest is the policy argument rather than the technical scenario analysis.

2. How the Documents Relate

The three documents form a coherent argument that operates at three levels simultaneously.

The Child Document establishes the threat. It answers the question: What does a military action against Oil and Natural Gas infrastructure actually look like, and what are the consequences? Without this foundation, the policy argument is abstract. With it, the policy argument is grounded in specific scenarios, specific consequence envelopes, and specific recovery timelines that make the cost of inaction concrete.

The LNG Use Case extends the argument to the alliance level. It answers the question: why does this matter beyond the domestic U.S. context? The LNG terminal is the point where the domestic energy security argument becomes an allied energy security argument — where the threat to U.S. infrastructure becomes a threat to European heating supply and NATO military

logistics. The Use Case is the document that makes the argument legible to NATO planners and European government officials who are not embedded in the U.S. regulatory debate.

The Policy Companion translates the threat into action. It answers the question: What do we do about it? It takes the threat established in the Child Document and the alliance-level stakes demonstrated in the Use Case. It converts them into a specific legislative and regulatory ask, with a specific pathway, specific actors, and specific asks for each.

The three documents can be transmitted together as a package or individually to audiences whose primary interest corresponds to one level of the argument. A congressional staff member who receives only the Policy Companion has everything needed to understand the legislative ask. A NATO energy security planner who receives only the LNG Use Case has everything needed to understand the alliance-level risk. A facility security officer who receives only the Child Document has everything needed to understand the threat scenarios and the gaps. The introduction document — this document — provides the context that allows recipients to navigate the set intelligently.

3. The Action Agenda

Four audiences need to act on the gap this series identifies to be closed. Each has a different role, a different ask, and a different timeline. This section addresses each directly.

3.1 Trade Association Leadership

Kimberly Denbow at AGA and her peers at INGAA, AXPC, IPAA, NGSA, and API are the coalition without which no structural changes occur. The trade associations are the sector's voice to Congress and to the regulators. They are the organizations with the sustained relationships, the technical expertise, and the institutional credibility to lead a mandatory standards initiative in a way that produces workable standards rather than regulatory burdens imposed without operational input.

The ask to trade association leadership is threefold.

First, read the Policy Companion. Specifically, read Section 1.0 — the CIKR logical predicate — and ask whether the argument is answerable. It is not. The government has already determined that mandatory standards are logically warranted. The trade associations' choice is not whether mandatory standards come — it is whether they arrive through proactive engagement or through reactive imposition after the next Colonial Pipeline. The Colonial Pipeline cybersecurity Security Directives are the proof of what reactive imposition looks like. The trade associations should not repeat that experience for physical security.

Second, assess the three-part offer in Section 6.5 of the Policy Companion — liability protection, cost recovery, and collaborative development — and determine whether those three elements, secured as conditions of trade association support, constitute an acceptable basis for engagement.

This is the business case that can be taken to member companies. It converts a regulatory imposition into a negotiated framework. The answer to each element should be yes, because the alternatives — tort liability without a compliance defense, compliance costs without rate recovery, standards written without industry input — are materially worse.

Third, initiate the coalition. AGA cannot do this alone and should not try. The coalition needs INGAA because it represents the midstream operators whose compressor-station infrastructure is the primary target in Scenario A. It needs AXPC because AXPC represents the upstream producers whose gathering infrastructure feeds that system. It needs IPAA because IPAA represents the broader independent community that constitutes the majority of the operator population. It needs NGSA because NGSA's supply chain relationships are central to the recovery architecture. A coalition that does not include all six major trade associations is one that the adversary can still exploit through the operators it does not cover.

The window for proactive engagement is open. It closes with the next significant physical security incident against Oil and Natural Gas infrastructure. That incident may be months away or years away. It is not decades away. The trade associations at the drafting table when mandatory standards are written will have produced standards that their members can live with. The trade associations that are at the comment table after crisis conditions have driven emergency directives will have received whatever the regulator wrote under pressure.

3.2 Member Companies of the Trades

The trade associations answer to their member companies. A trade association that supports mandatory standards without member-company backing is committing to something it cannot deliver on. Member company engagement is therefore not downstream of trade association leadership — it is the foundation on which it rests.

The ask to member companies has three components, each corresponding to a different organizational function.

The ask to security and operations teams is to read the Child Document and conduct an internal gap assessment. Specifically: map your highest-consequence facilities against the Tier 1, 2, and 3 asset classification framework in Section 7.1 of the Policy Companion. Identify which of your facilities would be covered under a mandatory standards regime equivalent to the one proposed. Assess your current physical security posture against the threat scenarios in Sections 4 and 5 of the Child Document. The gap between your current posture and the design-basis threat is the compliance investment your organization would incur under mandatory standards. Understanding that gap before legislation arrives is the difference between a managed transition and a reactive scramble.

The ask to legal and risk management teams is to read Section 6.5.1 of the Policy Companion on liability protection and assess the current exposure under the voluntary framework. An operator that fully complies with voluntary guidelines has no legally cognizable compliance defense in

litigation following a security incident, because voluntary compliance is not a standard of care. An operator that implements mandatory standards, verified by a third party, does. The question for legal and risk management is not whether liability protection is worth having — it obviously is — but whether the organization is currently positioned to benefit from it if mandatory standards are enacted, or whether internal security investment decisions made under the voluntary framework have left gaps that a third-party verifier would identify.

The ask to the finance and regulatory affairs teams is to read Section 6.5.2 on cost recovery and assess the rate case implications for regulated entities within your organization. Interstate natural gas pipeline operations subject to FERC jurisdiction have a pathway to recovering mandatory security compliance costs through regulated rates. This pathway does not exist under the voluntary framework, where security investments are borne entirely by shareholders. The compliance cost analysis for a mandatory standards regime looks materially different when rate-recoverable costs are separated from non-recoverable costs. Finance teams that have not modeled this distinction are making compliance cost estimates that overstate the net burden on shareholders.

The member company action that matters most, however, is the signal sent to trade association leadership. A trade association executive who hears from member companies that they support proactive engagement in mandatory standards development — on the terms of the three-part offer — has the political standing to lead the coalition. A trade association executive who hears nothing from member companies, or who hears opposition, cannot move. The member companies are the constituency. Their voice determines whether trade association leadership on this issue is politically sustainable.

3.3 Multinational Company Leadership

The sector's large integrated companies — ExxonMobil, Chevron, Shell, BP, TotalEnergies, Equinor, and their peers — have a pathway to action that does not require waiting for legislation. They are already operating in multiple regulatory jurisdictions. They are already subject to the EU's NIS2 Directive requirements in European member states. They already have corporate security standards that, in many cases, exceed the TSA's voluntary guidelines. They have the financial scale to implement mandatory-equivalent standards without rate recovery, and the corporate governance infrastructure to do so systematically across their asset portfolios.

The ask to multinational company leadership is to move ahead of the legislative curve on three specific fronts.

The first front is internal harmonization. A company that operates a U.S. LNG export terminal under voluntary TSA guidelines and a European LNG import terminal under NIS2 mandatory requirements is maintaining a two-tier security standard within the same corporate enterprise. That asymmetry is a governance risk, an insurance risk, and an investor relations risk. The internal harmonization calls for bringing U.S. asset security standards up to the higher of the two

tiers — effectively implementing mandatory-equivalent standards at U.S. facilities as a matter of corporate policy rather than waiting for regulatory compulsion.

The second front is public commitment. A multinational that publicly commits to meeting CIP-014-equivalent physical security standards at its U.S. Oil and Natural Gas assets — framed as alignment with European regulatory obligations and NATO energy security commitments — creates three simultaneous effects: it differentiates the company in the market for European government supply contracts and long-term buyer relationships; it creates investor relations value by demonstrating a security posture that voluntary-framework competitors cannot verify; and it establishes a market precedent that puts pressure on the broader operator population to follow. The first large company to make this commitment publicly sets the terms for the conversation that follows.

The third front is the European government-NATO relationship channel. Multinational companies with significant European operations have direct relationships with EU energy security bodies, NATO member-state governments, and the multilateral institutions that manage the post-Nord Stream energy supply architecture. Those relationships provide a channel for the physical security standards argument that no domestic regulatory debate can replicate. A multinational that communicates its U.S. physical security commitment to European government counterparties — and frames it as part of its alliance-level supply reliability guarantee — advances the allied energy security argument in a way that strengthens both the commercial relationship and the political argument for mandatory U.S. standards.

The multinational pathway and the trade association pathway are not alternatives. They are complementary tracks that create convergent pressure toward the same outcome. The multinationals provide the near-term market signal; the trade associations provide the legislative vehicle. Together, they create the conditions in which mandatory standards become a question of when, not whether.

3.4 Congress, Senate, and the Regulators

The legislative and regulatory audiences receive the Policy Companion as their primary document, with the Child Document available as technical support. The framing for each is different.

3.4.1 Congressional Staff

Congressional staff are the mechanics of legislation. They need three things: a clear argument, a legislative vehicle, and a political narrative that survives the anti-regulation environment of the current Congress.

The clear argument is in Section 1.0 of the Policy Companion: the CIKR logical predicate. The government has already determined that this sector's disruption has national security consequences. Voluntary guidelines are inconsistent with that determination. The question before

Congress is not whether to regulate — the government’s own designation has already answered that — but why it has not done so and what it will take to make it happen. That is an argument that does not require the staffer to advocate for more regulation in general. It requires only intellectual consistency with the government’s own stated position.

The legislative vehicle is the National Defense Authorization Act. The NDAA is the right vehicle because it frames mandatory Oil and Natural Gas physical security standards as a defense measure — necessary to support deterrence against state-level adversaries, protect NATO alliance energy supply commitments, and close a national security gap that the voluntary framework has left open. That framing changes the committee jurisdiction, the political valence, and the available coalition. It is not a regulatory bill; it is a defense bill. The distinction matters enormously in the current congressional environment.

The political narrative is Colonial Pipeline, Nord Stream, and the CIKR designation. Colonial Pipeline is the domestic proof of concept: a voluntary framework led to inconsistent implementation; a single incident exposed the gap; and the regulatory response was emergency directives written without industry input. Nord Stream is the international proof of concept: state actors are willing to destroy critical energy infrastructure on a strategic scale. The CIKR designation is the government’s own acknowledgment that the consequences matter at the national level. Put those three elements together, and the political narrative writes itself: the government said this sector was critical. An adversary proved it was a target. The voluntary framework proved it was undefended. The NDAA is the vehicle to close that gap before the domestic equivalent of Nord Stream chooses us.

The ask to congressional staff is to identify the NDAA provision sponsors within their respective caucuses, brief them on the CIKR logical predicate argument, and open the dialogue with TSA and FERC staff about the collaborative development process authorization that a statutory provision should include.

3.4.2 Legislators

The ask to legislators — House members and Senators with jurisdiction over energy, homeland security, or defense — is narrower and more direct than the ask to their staff. Legislators need the constituent framing, the cost structure, and the allied assurance argument.

The constituent framing is energy reliability and economic security. The disruption scenarios in the Child Document — compressor station attacks causing heating fuel shortages during winter, refinery attacks causing fuel price spikes, pipeline ruptures causing electric generation outages — affect constituents in every district in the country. The framing is not abstract national security; it is whether the heat stays on and whether the gas pump works. That is a constituent-facing argument that crosses party lines.

The cost structure is favorable because the three-part offer in the Policy Companion distributes compliance costs appropriately. Liability protection reduces the litigation exposure that would

otherwise generate opposition from the business community. Cost recovery through rate cases means that compliance costs for regulated pipeline operators are passed on to ratepayers rather than concentrated on shareholders, addressing the corporate opposition argument without requiring taxpayer appropriations. Collaborative development means that the standards are written with industry expertise rather than imposed by regulatory fiat, which addresses the anti-regulation argument. The result is a mandatory standards framework that does not require a government spending bill, does not impose unrecoverable costs on industry, and produces standards that the industry helped write.

The allied assurance argument is increasingly available as a political asset. The United States made explicit commitments to European allies to replace Russian pipeline gas with U.S. LNG supply after 2022. Those commitments are commercially real — they are embodied in long-term supply contracts between U.S. exporters and European buyers. A U.S. LNG export terminal that operates without mandatory physical security standards is a supply commitment that the United States cannot guarantee in the event of a military action or an attack on that terminal. European allies know this. The argument to legislators is that mandatory standards for U.S. LNG export infrastructure are part of meeting the commitments that the United States has already made to NATO partners.

3.4.3 TSA and FERC

The regulators are the most important near-term audience because they do not need to wait for legislation to begin. TSA has existing authority under 49 U.S.C. § 114 to issue Security Directives for pipeline physical security — the same authority it exercised for cybersecurity following the Colonial Pipeline incident. FERC has existing authority over interstate natural gas pipelines and LNG terminals under the Natural Gas Act of 1938 (NGA), and has existing precedent for recognizing mandatory compliance costs as rate-recoverable for electricity sector CIP standards. Neither agency requires new legislation to begin the engagement that a collaborative standards development process requires.

The ask to TSA is to signal publicly that the physical security voluntary framework is under review in light of the demonstrated threats — Nord Stream, Ukraine, the Houthis' maritime campaign — and that the agency is prepared to work with the sector's trade associations on a collaborative standards-development process. That signal, even without formal rulemaking, changes the dynamic: it tells the trade associations that the regulatory environment is moving and that early engagement is more valuable than late engagement. It also tells Congress that the agency is capable of acting and is seeking the authority to act more formally.

The ask to FERC is to issue a policy statement confirming that mandatory physical security compliance costs for interstate natural gas pipeline operators subject to FERC jurisdiction are prudent and recoverable through the rate-making process. This policy statement does not require legislation. It requires a Commission decision to extend the existing precedent for CIP compliance cost recovery in the electricity sector to the analogous situation in the natural gas

sector. That extension is legally straightforward and operationally important: it removes the single largest financial barrier to trade association support for mandatory standards, which is the concern that compliance costs will be borne entirely by shareholders without rate recovery.

Together, a TSA signal and a FERC policy statement change the landscape for the trade association coalition before a single line of legislation is drafted. They demonstrate that the regulatory environment is moving in the direction that the Policy Companion’s argument requires, that the three-part offer is not contingent on legislation alone, and that the window for proactive engagement is measurably open.

4. Open Questions and Work in Progress

The three documents in this series provide a substantial analytical foundation, but they do not constitute the complete picture. Several significant pieces remain to be developed, and their development depends on the engagement that this transmission initiates.

The physical consequence companion to the LNG Use Case. The LNG Use Case deliberately sets aside the BLEVE and vapor cloud explosion consequence envelope to focus on the strategic supply chain argument. That physical consequence analysis — the thermal radiation radii, the blast overpressure envelopes, the community evacuation implications of a major LNG release under military-action conditions — is a necessary companion document that has not yet been drafted. It is the document that USCG and emergency management audiences need, and it should be developed in coordination with LNG terminal operators who have the process engineering expertise to validate the consequence modeling.

The deterrence legislative document’s next steps. The Policy Companion makes the case for mandatory standards and identifies the legislative vehicles. It does not draft statutory language. The next step in the legislative track is translating the Policy Companion’s framework into specific amendment language for the NDAA provision and specific regulatory guidance language for the TSA and FERC actions. That work requires legal drafting expertise and congressional staff engagement that goes beyond what the analytical documents can provide.

The ONG Child Document and LNG Use Case are pending revisions. Both documents have identified revision items: the CIPAC termination acknowledgment, the full trade association inventory, and the softening of the ONG SCC language. These revisions are flagged and will be incorporated in the next revision cycle of each document. They do not affect the core analytical content.

National child documents for additional sectors. The DBA-MA framework is designed to be extended across sectors. The electricity, water, and transportation sectors all present analogous military-action threat profiles that the DBA-MA methodology can address. Those extensions are in the queue but have not been initiated.

Allied coordination exercises. The LNG Use Case identifies the absence of a NATO energy security exercise in military-action scenarios as a material gap. Initiating that exercise — through the relevant NATO bodies and European Commission energy security channels — requires contacts and relationships that the trade associations and multinational companies are better positioned to activate than any individual author. If Kimberly or her peers at the other trade associations have relationships with NATO energy security staff or European Commission energy officials, that is the most valuable near-term contribution they can make to the allied coordination gap.

Appendix: Document Series Summary

Document	Level	Version	Primary Audience
DBA-MA Oil and Natural Gas Sector Child Document	L2	V1.1	Facility security officers, exercise designers, sector risk managers, and government emergency planners
DBA-MA Use Case: LNG Terminal Attack	L3	V1.0	NATO energy planners, European government officials, multinational executives, and maritime security practitioners
Deterrence Without Standards: The Case for Mandatory Physical Security Requirements	L2-P	V1.1	Trade association leadership, congressional staff, TSA, FERC, multinational C-suite, and government affairs
Introduction and Transmission Guide (this document)	Series Overview	V1.0	Kimberly Denbow, AGA, and peers across the ONG sector trade associations

Documents in this series are unclassified and distributable to domestic and international audiences. All analytical content is derived from publicly available sources and established safety analysis methodology. The purpose of these documents is to enable protection, not to provide targeting information.

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